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Problem Set 4

Problem 1

Small correlation coefficient : data variability similar in every direction
Big " " : data variability ($\Leftrightarrow$ magnitude of spread) significantly higher in one

The diagonalisation try to maximisize the scalling operation /stretching/compression, so correlation results in a value in the diagonal matrix higher than the other because the data will strongly spread in a direction.

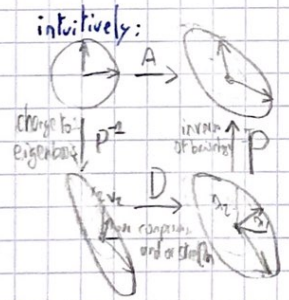
Problem 2

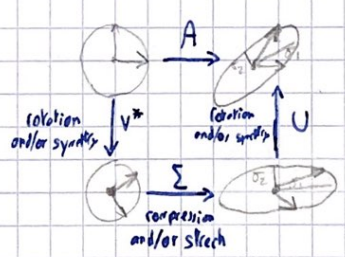
Eigen Value Decomposition Vs Singular Value Decomposition

$$\underset{n\times n}{A} = \underset{n\times n}{P}\,\underset{n\times n}{D}\,\underset{n\times n}{P^{-1}} \qquad\qquad \underset{m\times n}{A} = \underset{m\times m}{U}\,\underset{m\times n}{\Sigma}\,\underset{n\times n}{V^*}$$

A is a square matrix	A can be rectangular
P is inversible but not necessarily orthogonal	U, V orthogonal but not necessarily inverse of each other.
D can be complex, negative	Σ real and non negative

intuitively:





$$Ax = \lambda x \qquad\qquad \sigma x = Ay$$

A is described by the vectors that are scaled but doesn't rotate

A is described by the vectors that are scaled by the same amount

$$\left(\begin{aligned} AA^T = BB^T &= U\Sigma V^T V \Sigma U = U\Sigma^2 U^T \Rightarrow \\ &= PDP^{-1} \qquad \sigma = \sqrt{\lambda} \end{aligned} \right)$$

They give the same decomposition if : A has orthonormal eigenvectors and positive eigenvalues $\Leftrightarrow$ A is symetric positive-semi-definite $\Leftrightarrow \exists B \in \mathbb{R}^n$ such that $A = B^T B$

Covariance Matrices are always positive semi-definite
$(\Leftarrow x^T K_{zz} x = x^T E[(z-\bar{z})(z-\bar{z})^T] x = E[x^T(z-\bar{z})(z-\bar{z})^T x] = E[Y^2] = \sigma_Y^2 \geq 0)$

Problem 3

$$z \sim p_z(z) = \mathcal{N}(\bar{z}, K_{zz}) = \frac{1}{\sqrt{(2\pi)^N |\det K_{zz}|}} e^{\left(-\frac{1}{2}(z-\bar{z})^T K_{zz}^{-1}(z-\bar{z})\right)}$$

$$p_x(x) = \frac{p_x(A^{-1}x)}{|\det A|} = |\det A|^{-1} \cdot \left(\frac{1}{\sqrt{(2\pi)^N |\det K_{zz}|}} e^{\left(-\frac{1}{2}(z-\bar{z})^T K_{zz}^{-1}(z-\bar{z})\right)}\right)$$

$$= \frac{1}{|\det A|} \cdot \frac{1}{\sqrt{(2\pi)^N |\det K_{zz}|}} e^{-\frac{1}{2}(A^{-1}x - \bar{z})^T K_{zz}^{-1}(A^{-1}x - \bar{z})}$$

$$= \frac{1}{\sqrt{(2\pi)^N |\det A|^2 |\det K_{zz}|}} e^{-\frac{1}{2}(x - A\bar{z})^T A^{-T} K_{zz}^{-1} A^{-1}(x - A\bar{z})}$$

$$= \frac{1}{\sqrt{(2\pi)^N |\det A K_{zz} A^T|}} e^{-\frac{1}{2}(x - A\bar{z})^T (A K_{zz} A^T)^{-1}(x - A\bar{z})}$$

$\Rightarrow$ by identification $\begin{cases} \bar{x} = A\bar{z} \\ K_{xx} = A K_{zz} A^T \end{cases}$

Alternatively

$$\bar{x} = E[x] = E[Az] = A E[z] = A\bar{z}$$

$$K_{xx} = E\left[(x - \bar{x})(x - \bar{x})^T\right] = E\left[(Az - A\bar{z})(Az - A\bar{z})^T\right]$$

$$= E\left[A(z - \bar{z})(z - \bar{z})^T A^T\right] = A E\left[(z - \bar{z})(z - \bar{z})^T\right] A^T$$

$$= A K_{zz} A^T$$

$$\Rightarrow p_x(x) = \frac{1}{\sqrt{(2\pi)^N |\det A K_{zz} A^T|}} e^{-\frac{1}{2}(x - A\bar{z})^T (A K_{zz} A^T)^{-1}(x - A\bar{z})}$$

Problem 4 Hamming weight on a large sample has the same shape as the binomial (/Gaussian) pmf

the smaller θ is the smaller the mean gets. The closer θ is to 0.5, the higher the variance is

The bigger N is, the peakier the shape gets ($\Leftarrow$ variance)

// The less probable a sequence will end on an exact number and so the lower the peaks ($\theta = \frac{1}{3}$ a bit higher than $\theta = \frac{1}{4}$ because of the smaller variance)